

Predictive Placement And Routing Of Charging Station For Electric Vehicles

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Abstract

The utilization of EVs can be considered as a breakthrough of the 21st century because such vehicles solve many problems related to pollution occurred because of utilization of fossil fuels and provide drivers with more comfort compared to traditional cars. Whereas, conventional vehicles entail expensive re-fueling, with EVs, one could charge them under 100-200 rupees to full charge. As the demand of EVs grows day by day, installation of charging stations and finding the nearest station in emergent condition is important. Both challenges are addressed in this paper through employing of the machine learning algorithms. The routing model of our application employs Dijkstra's algorithm to direct the nearest charging station with an ideal route to the user's location based on their guild, estimated destination, battery level, and charging time.

Also, the present work utilises a predictive placement model that has a random forest regressor to make fu-

ture estimations on the most suitable places for new EV charging stations to be located. The above model was developed using relevant features within population density, traffic congestion, level of urbanization, and distance to highways amongst others. During our experiments that were carried out in Delhi, India, the suggested placement prediction model was quite successful. For latitude prediction, it obtained a Mean Squared Error (MSE) of 0.04, Root Mean Squared Error (RMSE) of 0.21, Mean Absolute Error (MAE) of 0.16, and an R-squared (R2) score of -0.19. For longitude prediction, it achieved an MSE of 0.05, RMSE of 0.22, MAE of 0.16, and an R-squared (R2) score of -0.16. Based on these outcomes, it can be concluded that the proposed model is efficient in determining the areas relevant to EV charging equipment installation, thereby enhancing the comfort level of the overall EV system. Thus, the work presented in this paper suggests a solution based on the combination of Dijkstra’s algorithm and random forest regressor to guarantee that EV users get a fast and efficient charge to contribute to the achievement of the goal to promote the utilization of electric vehicles.

1 Introduction

The acceleration of Electric Vehicles is a critical transition towards clean mobility powered by the must of act on climate change and energy security. However, the feasibility and social acceptance of Electric Vehicles in our everyday life greatly depend on the development of charging stations for these vehicles. The topic "Pre-

dictive Placement and Routing of Charging Stations for Electric Vehicles” addresses a fundamental challenge in this evolving landscape: improving logistics to position charging stations and demand necessary for EVs demand well.

Optimal deployment and interconnection of charging points are crucial to tackle range anxiety, one of the major concerns of drivers reluctant to switch to EVs. This can be solved by the creation of production of charging stations which can result in range anxiety in the sense that the owners of EVs can be sure that they have enough charge to reach their destination. The traditional approach for identifying charging station places is simply being reactive and putting stations where cars currently go and individuals are calling for. However these methods can result in inefficient resource use and also make it harder to access for example as the usage patterns of EVs change.

Predictive analytics provides just the kind of long-term perspective that decision-makers in this sector require – it is a data-driven approach that aims to forecast the future demand and optimize the location and connections for charging stations. Using methods like neural network forecasting models, the owners can predict the criticality of locations beforehand and then decide where new charging stations need to be established. Also, the algorithms for routing can guarantee EV drivers to quick and stable connect to the charging stations along the routes, which will also be of great convenience and feasibility for EV drivers.

This research paper employ predictive analytics in place-

ment and routing EV charging stations including methodologies, benefits and challenges of adopting this approach. This paper studies several real world case studies and uses data modeling methods to further understand how predictive placement and routing algorithms can be used to improve the EV charging infrastructure by creating a more sustainable transportation system.

2 Literature Survey

The development of Batteries [2], [3] and techniques makes the implementation of Battery electric buses (BEBs) done easily for sustainable mobility. Still, relative to the conventional diesel or Compressed Natural Gas (CNG) buses, the constraints that BEBs have include; battery capacity, high capital costs, limited driving range, and compulsory charging equipment.

The prior studies done on diesel or CNG buses mainly included the design of transit service (For instance: coverage zones, time-table) [27], [28] and the other bus characteristics (For instance: size of fleet, capacity level of buses etc) [29]. These aspects still apply to BEB systems because BEBs and diesel/CNG buses fulfil similar objectives of reducing traffic congestion, enhancing people’s access to employment opportunities, and reducing the cost of transportation to the public. However, to support OpenFlow, BEBs have extra limitations that have to be considered when deploying the solutions.

For instance, in the study by Spasovic et al [27] on optimal transit coverage, such issues as the distance of the transit routes from the CBD and how they branch out

to the surrounding less populated suburbs, the routes' frequency or gaps, fares, etc are important; for BEBs, factors like the possible charging points, charging time, range among others require attention. As in [28], where multiple bus stop spacing optimization in urban areas is presented, the spacing between them charging sites has to be within the range of the BEB's drive distance.

However, in the light of technological content's being immature and capital intensive, transit electrification has emerged as an important factor for developing smart cities [30]. BEBs motivate innovation in some areas: charging infrastructure, batteries, and automobile control [31]. There is also the possibility of BEBs' integration into such innovations as telecommunications, Internet of Things, and cloud technologies [32]. For instance, [31] illustrated a merged structure with the in-vehicle and cloud modules to execute rules and policies based on live events, including verifying the attainability of the nearest charge stations in relation to the state and efficiency of the buses.

Earlier some researches have focused on location of charging facilities of private or alternatively fueled cars, regarding driving traits and battery consumption assessment [12], [14]. For instance, [6] uses a model to decide the policy on the depot location of charging facilities required at bus stops while minimizing the total cost of the investment. The efficiency of this model was also confirmed through testing it on the data of another make-believe company. Other similar works such as [15, 33] also present siting patterns that will reduce capital expenses. Regarding the fleet replacement strate-

gies, they have been discussed in [20], but these often involve certain assumptions, for instance, optimizing only the costs or neglecting the fact that the BEB system is rather sensitive to the spatiotemporal context. BEBs have limits on the range and have to be recharged from time to time and this has to be feasible from already established routes and schedules when replacing diesel or CNG buses. Such considerations are related to on-route charging time windows, charging conflict, and bus traces. Also, majority of the research works conclude assuming full fleet replacement with BEBs; however, transit agencies may decide on phased replacements. For example, [6] did not at all take into consideration actual bus schedules, [13] only looked into certain routes, and [20] failed to demographically consider on-route and in-depot charging locations.

The literature on BEBs also speaks on the level of environmental reductions from conversion from normal buses. There are several LCA studies that have been conducted in various researches, for instance; [22], [34]–[37] as well as the examination of emissions of other air pollutants [38]. For example, [34] the charging intervals was done based on CO₂ emission and price of electricity and [35] they assessed the LCEI of city buses with varying levels of electrification decentralisation, charging solutions, and fuels. Comparing the expected variations in energy demand and greenhouse gases emissions, as well as in the pollution of the air were analyzed in an empirical assessment at Rome, Italy [38].

Concerning the third research gap, the above analy-

sis revealed that most of the research works done in this area is specifically concentrated on the parameter of cost or environmental effects of BEB deployment social equity factor in the public transit planning has remained uncovered. More important, transit-oriented populations are highly associated with the high concentrations of air pollution. For these reasons it is preferable to use BEBs rather than diesel or CNG buses in these neighborhoods because in social equity of the community and the environment [39].

To sum, some previous studies have taken simple assumptions to investigate the charging station location and fleet replacement, but these studies essentially do not guide the transit agencies to find how, step by step and complexly, they have to deploy the BEBs. The study will therefore aim at proffering bi-objective model to help transit agencies distribute BEBs' spatial-temporal density with respect to capital investment and environmental equity criteria as influenced by the operational characteristics of the BEB system per geographical mile and time of the day.

3 Methodology

3.1 Data collection

The routing model requires a dataset of all the locations where the EV charging stations are placed. For our project, we had taken the list of all EV stations present in Delhi, India. The dataset was present on the government's official website. For the predicting model, it

requires a huge dataset with the coordinates of all the locations present in an area along with the population density, traffic level, urbanization level and distance to highway. These were the attributes that we have currently used to build our model. Since the size of this dataset is very large and is very vast, it wasn't available anywhere. With the help of a random variable, we generated a dummy dataset of a 100 places in Delhi, with random values of all the attributes, just to train the model. These 100 places are the hotspot locations of Delhi. After the model had been trained, we replaced the dummy values of the top 5 locations with their actual values just to check if the model worked, and it produced commendable outputs.

3.2 Graphing

The coordinates of all the locations of the EV stations were then added with the help of GeoCode, an extension on Google Sheets. These coordinates were then plotted on a map by a python library named OSMnx library to obtain OpenStreetMap data for Delhi. It then creates a road network graph using the obtained data and projects the graph to Universal Transverse Mercator (UTM) for accurate distance calculations. The charging stations are added as nodes in this graph

3.3 Model

3.3.1 Routing

The model takes inputs: Current Location, Destination Location, Battery Charge and Charging Time. We as-

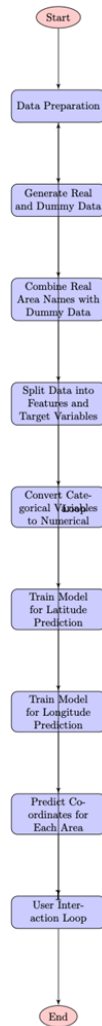


Figure 1: Flow-Chart



Figure 2: Delhi EV Charging Stations

sume that 60km of range is added when the battery is charged for 1 hour. We calculate the range of vehicle by seeing the battery charge left and the charging time. the main motive is to check which charging station is within that range and nearest to the user. The user enters their current location and their destination location as well. These locations are then converted into coordinates with the help of GeoCode. Now with all the inputs present, we use Djikstra's Algorithm to find out the shortest path between the starting and end point. If the charging range is less than the distance between these two points, the model directs the vehicle to the nearest charging station for it to refuel and then go ahead.

Djiksta's Algorithm Psuedo Code

Djikstra(Graph G, Vertex S)



Figure 3: Delhi EV Charging Stations Possible Locations

```

Initialize(G,S)
PriorityQueue minQ = all vertices in V:
while(minQ!=phi) do
Vertex u = ExtractMin(minQ);
for(each v belongs to minQ such that (u,v) belongs to E):
Relax(u,v)
end for
end while

```

3.3.2 Predictive Placement

Our work aims to predict the specified places from user reviews for recommend organization of forecast accurate predicted locations mainly on type EV charging station includes diverse set of place names. The concept is to find public places where people come and gather from hotels, restaurants, hospitals etc., anywhere you could place a station in an accessible form. Details such as latitude, longitude, population density and highway distance, degree of urbanization and volume of traffic; the type of area and statistics on electric vehicle (EV) usage are part of the dataset. These features collectively influence the optimal deployment of EV charging infrastructure within urban environments.

Employing this vast set of data, and with the objective of predicting the latitude and longitude coordinates, the proposed model is a Random Forest regression. The availability of the dataset was in the form of a single set; as such, splitting was carried out to infer the ability of the model in terms of accuracy and

performance. In Random Forest regression model, several decision trees are employed and every tree gives value prediction based on different data samples and attributes used in the tree model. As indicated, the conformal regression-based R-ensemble approach yields well-calibrated and non-overfitting probabilistic forecasts of continuous-valued variables by not over-relying on any one data point or feature.

On successful training, this model runs online in real-time and can make available potential EV charging station location for user queries. Users enter the name of the desired area in the system upon arrival. It employs the developed model to predict the coordinates of the required EV charging stations. In essence, many potential locations are created by the Random Forest model through iteration of the prediction process. The iterative approach eventually gives the stakeholders a detailed list of various possible sites for EV charging stations, which enhances decision making.

4 Experiment

4.1 Working of the Routing model

After taking the dataset of all the electric vehicle charging stations in Delhi and plotting them, we find the nearest charging station for the user to help making their journey a bit easier and more relaxing. After running the model, we saw that the routing model directs the user to the nearest charging station and on comparing the user's geocode with other EV stations' location, the

location calculated by the model is the most accurate and correct.

4.2 Working of the Predictor Model

On taking into account all the attributes, the optimal location's coordinates are predicted with the help of random forest regressor. Since we have taken only the coordinates of the hotspot locations in Delhi, the model predicts a location which is no more than 5 km away from the current location. This is calculated by taking into account all the attributes of the starting location and the optimal place's location. The scores that have been calculated are Mean Squared Error(MSE), Root Mean Squared Error(RMSE), Mean Absolute Error(MAE), R-squared(R2) score.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y - y_i)^2 \quad (1)$$

5 Future Scope

The prediction model, as we mentioned before, has been trained on a dummy dataset as no dataset had what it requires, i.e. location of all points in a city or town. The dataset also requires the values of other attributes on which the model has been trained. The model is ready and working and as soon as a dataset like this, with all the necessary values is made or is available anywhere, this model is ready to work on large scale projects as well. Updates that can be made is to make the model

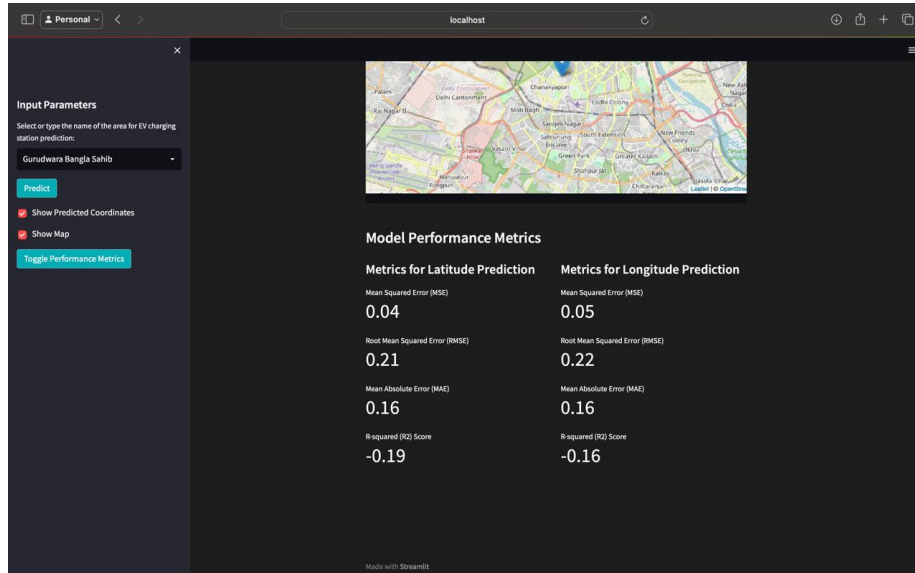


Figure 4: Output with values of errors and scores

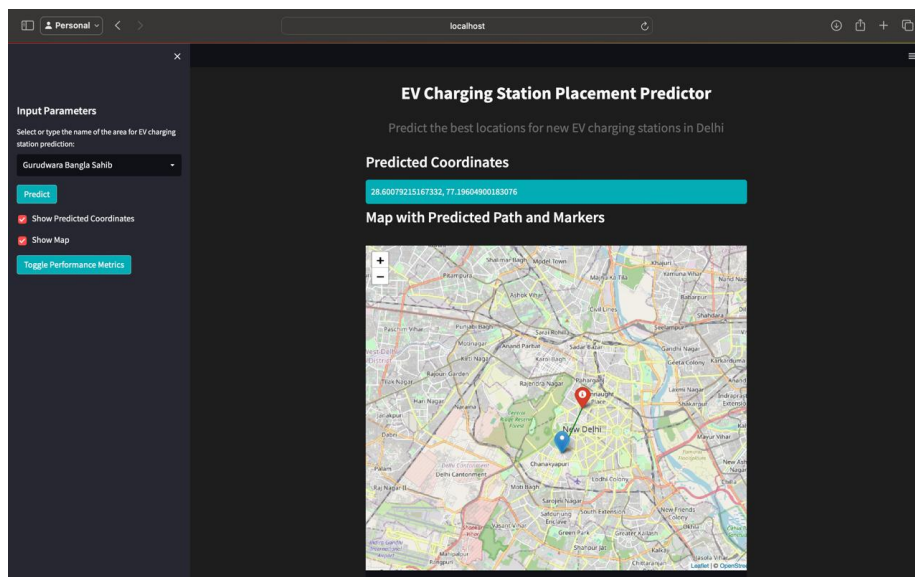


Figure 5: Plotting of the predicted location on map

more optimizing such that it does not predict 2 charging stations adjacent to each other. For example, place 1 gives output 1 and place 2 gives output 2. if these 2 outputs are adjacent to each other, then it will be just a waste of resource.

6 Conclusion

EV vehicles are the future and to prevent dealing with the problem of scarcity of EV stations later on, we should be prepared for it now only. Both the models, routing and predictive placement are a necessity and will benefit the people to a huge extent. The findings obtained in this study show that the charging infrastructure can be made more efficient and the users can have a better experience. This paper has also suggested ideas on how to alleviate congestion at charging points by placing the charging points strategically and optimizing the routes that the EV users take in order to reduce time spent at charging points and hence increasing the practicality of widespread adoption of EVs.

This work contributes a holistic model for the optimal location and connection of EV charging stations. The proposed methods do not only solve existing problems but also foresee future requirements, thus providing a scalable and sustainable solution for EVs. Further research work can seek to improve this model by including other factors that may include availability of renewable energy sources, the economic climate, and government policies in an effort to optimize the performance of the EV charging station planning model. In conclu-

sion, there is much that can be learned from our study to inform the design of future transportation systems that rely on predictive and data-analytic approaches, to maximize the efficacy and minimize the environmental impact of transportation solutions.

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